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Routing Protocol for Wireless Sensor Networks using an Energy Efficient Binary Sculptor Artificial Protozoa Optimization Algorithm

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Abstract

Routing algorithms are effectively designed to decrease energy consumption and increase communication reliability to enhance the lifespan and performance of Wireless Sensor Networks (WSNs). Routing and clustering are important for sensor nodes to regulate energy consumption and allocate workload. However, individual nodes have limited power resources, and WSNs are dynamic and establish longer network lifetimes, which continues to be a major problem. To resolve these challenges, the Energy-Efficient Binary Sculptor Artificial Protozoa Optimization (EE-BSAPO) algorithm, a sophisticated routing protocol, is designed. The suggested protocol integrates the Binary Sculptor Artificial Protozoa Optimization method with an improved fuzzy logic-based clustering mechanism to enhance multi-hop routing and cluster formation. To enhance the cluster head selection and route discovery, based on distance, communication latency, trust value, and residual energy. Based on various simulations, EE-BSAPO significantly enhances network performance as compared to existing routing methods. In particular, its average end-to-end latency measured a very low 9.36 ms, its average throughput reached 75.24%, and its average network lifespan is 73.81%. The model is suitable for WSN because energy supplies are limited, based on its average energy efficiency of 72.32 J. Thus, the proposed work efficiently optimizes the energy efficiency, decreases communication overhead, and expands the network lifetime in WSN.

Keywords Fuzzy Logic System · EE-BSAPO · WSN · Routing Algorithm · Cluster Head · Energy Consumption · Communication Overload

1 Introduction

Wireless Sensor Networks (WSNs) are dispersed, self-organizing network architecture made up of many sensor nodes with wireless communication for local computing and environmental monitoring capabilities [1, 2]. WSN networks scalable and affordable option for real-time data collection in various applications such as smart homes, smart grid infrastructure, industrial automation, military surveillance, and habitat monitoring [3]. In a WSN, every sensor node is commonly compact by memory, compute power, and energy resources, all of which have an effect on the network's overall durability, scalability, and efficiency [4–9]. These nodes are function



independently in unreachable locations and hostile with their limited resources. As a result, these sensor nodes require strong communication protocols and efficient resource management [10, 11].

Since sensor nodes are normally energized by non-rechargeable batteries, energy consumption is biggest issues with WSNs [12]. Because in remote areas sensor nodes batteries cannot replaced, refilled, energy-conscious communication techniques must be developed [13–18]. Conventional clustering-based protocols with hierarchical designs are applied to reduce duplicate data transmission and distribute the burden across nodes [19]. This techniques are used in periodic cluster head (CH) rotation and randomized cluster formation to optimize energy efficiency [20]. However, CHs are utilized in long-distance communication and data aggregation, these methods result in an unequal allocation of energy [21]. Due to this, CHs use their energy more quickly, which affect network fragmentation and early node failures.

This study presents an Energy Efficient Binary Sculptor Artificial Protozoa-based Routing Protocol (EE-BSAPO) developed for WSNs to overcome these limitations. In addition, to achieve optimal cluster formation and CH selection, the designed work integrates an Improved Fuzzy-based System, for balanced energy use across the network. During the routing phase, most efficient communication paths are found using the EE-BSAPO method, because behaviour of Euglena protozoa and improved by sculptor-inspired modifications. It use of a binary optimization framework that is directed by a multi-objective fitness function depend on transmission delay, distance and residual energy. This makes it possible for the routing model to enhance energy consumption, ensure consistent delivery of data and dynamically adjust to shifting network topologies. Overall, by guaranteeing energy-efficient clustering and routing in WSN contexts, the designed protocol optimize network performance and lifetime.

1.1 Motivation

WSNs are important for communication and real-time monitoring of application, but their performance is severely limited by low processing power, low energy resources, and susceptibility to security risks. Conventional routing protocols use clustering and CH rotation to increase energy efficiency, but they are not flexible enough to change condition of network and are not effective in fight against malicious activity. Moreover, while choosing routing pathways, many optimization-based methods ignore trust and reliability considerations, which might risk to data and cause network instability. So, there is a need of a strong routing system that blends dependable and secure communication with energy economy. To improve network lifetime, secure data delivery, and balanced energy consumption across WSN nodes, proposed study presents Energy Efficient Binary Sculptor Artificial Protozoa Optimization (EE-BSAPO) protocol, which combines an Improved Fuzzy-based clustering mechanism with a trust-aware, multi-objective optimization strategy. The main contributions of the research are:

- For making cluster formation and cluster head selection, this paper introduced an Improved Fuzzy based System.
- For attaining an energy efficient, reliable and scalable data communication, routing is performed by proposing an Energy Efficient Binary Sculptor Artificial Protozoa Optimization (EE-BSAPO) algorithm.

Paper organization: Sect. 2 explains the related work along with the problem statement, Sect. 3 describes the proposed methodology with the necessary figures and equations. Section 4 illustrate the result and discussion. Section 5 represents the conclusion and future scope which the reference are shown in Sect. 6.

2 Related Works

A new approach called Deep Learning–Hybrid Energy-Efficient Distributed (DL-HEED) was presented by Juwaied et al. [22] and included a Graph Neural Network (GNN) into the clustering procedure. DL-HEED made intelligent, context-aware, and adaptive CH selection possible by utilizing the relational structure of WSNs and a

wide range of node and network characteristics, such as residual energy, node degree, spatial location, and signal intensity. Compared to conventional heuristic-based procedures, DL-HEED enables more flexible and energy-efficient cluster head selection by using the relational structure of WSNs through deep learning. Nevertheless, DL-HEED still has drawbacks, especially its high computational cost and dependence on large amounts of training data, which might impair real-time performance in WSNs with limited resources.

Suresh Kumar et al. [23] suggested energy-efficient and trust-aware routing architecture in which the Exponentially-Ant Lion Whale Optimization (E-ALWO) method was added to optimize data packet routing to target nodes. The Exponentially Weighted Moving Average (EWMA) approach integrated ALO and WOA to create the E-ALWO algorithm, which improves convergence speed and solution variety. This model uses CHs to carry out routing, and E-ALWO algorithm chooses CHs based on important performance metrics including communication latency and residual energy. However, the method ignores important trust-based parameters but it provide safe and dependable communication in the presence of hostile or compromised nodes, instead focusing exclusively on energy and latency metrics.

Cherappa et al. [24] used the Clustering and Multihop Routing Protocol (CMRP) for data transmission and combined K-medoids clustering with Adaptive Sailfish Optimization (ASFO) in WSN for effective cluster head selection. The K-ASFO approach is used to select the best cluster heads (CHs) based on performance from pool of qualified sensor nodes. The efficiency of this model is balance power consumption and prolonging network longevity was demonstrated depend on its performance indicators like throughput, residual energy, and time to first node death. After that, multihop routing is executed with the help of CMRP algorithm, which enables data flow over preset node-to-node communication channels. Nevertheless, in hostile or dynamic contexts, the model's absence of a thorough trust evaluation mechanism leaves it vulnerable to malevolent node activity and lowers the routing process's overall security.

Dogra et al. [25] solved the issues of traditional WSN routing methods by introducing Enhanced Smart-Energy-Efficient Routing Protocol (ESEERP) to increase connectivity and network longevity. For efficient Cluster Head (CH) selection, the protocol integrates a multi-objective optimization framework with focus on lowering energy usage and idle or inactive sensor nodes. Following the identification of the CHs, the Sailfish Optimizer (SFO) algorithm is applied to find the best routing path from the CH to the sink node, enabling energy-conscious data transfer. However, the protocol is flexible in vulnerable to malicious attacks or node penetration in CH selection and routing decision because of lack of trust and security.

In order to address power imbalance and inefficient routing, Natesan et al. [26] proposed energy-efficient clustering and routing protocol for WSNs use hybrid Mayfly–Aquila Optimization (MFA-AOA) algorithm. To select the best CH factors like energy level and closeness, Mayfly algorithm was used in this method. To improve power economy, data transmission reliability, Aquila Optimization Algorithm (AOA) was used to detect most efficient communication between chosen CH and base station. Nevertheless, the developed model limited in trust-aware features, but it is important for guaranteeing safe routing in hostile WSN settings.

A secure and energy-efficient routing system for Internet of Things applications running on heterogeneous WSNs was developed by Nagaraju et al. [27]. Multipath Link Routing Protocol (MLRP) was applied in proposed architecture to provide safe routing for the transfer of private IoT data among sensor nodes with various energy levels. To balance network load, improve power efficiency, and increase network lifetime Hybrid-based Threshold-sensitive Energy Efficient Network (H-TEEN) protocol is used. In addition, the Ubiquitous Data Storage Protocol (U-DSP) was linked to enhance data storage capabilities and facilitate the effective management of sensor-generated data. Adaptive trust evaluation during routing, which is important for identification of malicious nodes and guaranteeing safe data delivery in dynamic and hostile WSN environment, was not used in this model.

To generate multipath protocol referred as Multipath Particle Swarm Optimization Routing Protocol (MPSORP), Ghawy et al. [28] presented an optimization-based routing technique which use Particle Swarm Optimization (PSO) algorithm. This protocol is applied to address issues with high traffic volumes and unbalanced flow in WSN-based Internet of Things applications. MPSORP utilizing PSO algorithm to find many routing paths to

enhance network throughput and load distribution. However, lack of energy-awareness and trust measures reduce the path selection process's efficacy in guaranteeing secure communication and long-term network sustainability.

Akram et al. [29] adopted Machine learning (ML) based clustering and routing for WSN. The study has investigated the application of Q learning and K-means clustering for energy efficient routing. The ML based K-means clustering method group the nodes and cluster heads are selected based on residual energy. Then routing was made through Q-learning, which initialize the Q-values, hop nodes are chosen using greedy policy and updated Q-value using update rule. In this manner nodes are periodically re-clustered to balance the energy consumption. The challenges faced by this study include increased computational complexity during clustering and routing, high network congestion and collision.

Alanazi et al. [30] performed ML based routing for energy efficient communication in WSN. The framework has combined supervised learning with reinforcement learning for energy efficient data forwarding. The ML based supervised learning method extract features from sensor node and optimal path selection made using Q-learning. However the integration of ML with reinforcement has added computational overhead, this remains challenging for resource constrained applications due to limited memory, processing power and battery capacity.

Gupta et al. [31] introduced ML and glow worm swarm optimization algorithm for multi-hop routing in WSN. The framework performed multi hop routing by integrating network, radio and energy model, clustering made based on low energy adaptive clustering hierarchy (LEACH) and cluster heads are selected based on glow worm swarm optimization algorithm. Multi-hop routing done by Taylor cat-scalp swarm method, ML based k-means and hierarchical clustering. Although the optimization algorithm improved energy efficiency, clustering and optimization algorithm added additional energy consumption and scalability issues. Table 1 represents the consolidation table of existing studies.

2.1 Problem Statement

WSNs are applied in applications like environmental monitoring, industrial automation, healthcare and military operations. The sensor nodes intrinsic limitations such as their limited storage capacity and battery life, processing power, provide serious hurdles to enhance network performance and dependability. Conventional routing protocols are based on hierarchical clustering, which fall short of striking the best possible balance between network scalability and energy economy. Normally, to choose the CH protocols use probabilistic or static techniques, which result to unequal energy allocation and early node failures. Furthermore, a large number of optimization-based routing strategies now in use ignore the influence of trust and node behaviour in hostile contexts in favour of concentrating solely on reducing energy consumption and communication time. WSNs are susceptible to both external and internal security threats in absence of strong trust mechanism, which lead to packet loss, data manipulation, and decreased performance of network. Consequently, a thorough routing protocol that integrates trust-based decision-making in addition to optimizing energy consumption and communication efficiency is desperately needed to improve the general security, dependability and longevity of WSNs.

3 Proposed methodology

WSNs comprised several sensor nodes and are generally powered by small batteries. Thus, the nodes have limited energy capacity and leads node failure, reduce network lifetime and impacts overall reliability and performance. Therefore, this study goal is to design novel routing protocol for energy efficient to enhance lifetime of network connectivity. Initially, the system model is designed by deploying many sensor nodes in a WSN environment. After, sensor nodes are grouped into different clusters for improving scalability and reducing overall energy consumption. From each cluster group, one node is selected as a cluster head for an efficient data aggregation. Thus, for making cluster formation and cluster head selection, this paper introduced an Improved Fuzzy based System. The cluster heads collects data from its member nodes and send it directly to the base station. For attaining an

Table 1 Consolidation table

Reference	Technique	Limitations	Performance
Juwaied et al. [22]	DL-HEED	High computational cost and dependence on large amounts of training data, which might impair real-time performance in WSNs with limited resources	Average energy consumption, Packet delivery ratio, Average end-to-end delay, First Node Death (FND), Half Nodes Alive (HNA), and Last Node Death (LND)
Suresh Kumar et al. [23]	E-ALWO	Ignores important trust-based parameters that are necessary to provide safe and dependable communication in the presence of hostile or compromised sensor nodes	alive nodes number, FND, LND, time, Packet sent number
Cherappa et al. [24]	CMRP- ASFO	The model's absence of a thorough trust evaluation mechanism leaves it vulnerable to malevolent node activity	Time, Number of alive nodes
Dogra et al. [25]	ESEERP	The protocol's resilience in settings vulnerable to adversarial attacks or node penetration is limited by the lack of trust and security considerations in CH selection and routing decisions	packet delivery ratio energy consumption, jitter, packet delay, network lifetime, packet loss ratio throughput, buffer and occupancy
Natesan et al. [26].	MFA-AOA	The suggested model lacked trust-aware features, which are essential for guaranteeing safe routing in hostile WSN settings	Network Lifetime, Throughput, Energy Consumption, Residual Energy, Delay
Nagaraju et al. [27]	MLRP	The model is complex in nature	FND, HND and LND, residual energy
Ghawvy et al. [28]	MPSORP	The path selection process's efficacy in guaranteeing secure communication and long-term network sustainability was constrained by the lack of energy-awareness and trust measures.	Residual Energy, Delay, Throughput, trust
Akram et al. [29]	K-means Clustering and Q-learning	Increased computational complexity, high network congestion and collision.	Latency, Energy Consumption, Remaining Energy Level, Packet Reception Rate.
Alanazi et al. [30]	Supervised Learning	The performance was based on the quality of data, this reduced the accuracy of routing decision.	Packet delivery ratio, Residual energy
Gupta et al. [31]	LEACH, K-means and hierarchical clustering	The methodology added complexity due to combination of ML and optimization algorithm.	Accuracy, R2 value, Packet delivery ratio.

energy efficient, reliable and scalable data communication, routing is performed by proposing an EE-BSAPO algorithm. This algorithm selects the most optimal route as per the fitness parameters like distance, residual energy, minimum delay and maximum trust value. Figure 1 shows proposed work basic block diagram.

3.1 Assumptions

- The homogenous sensor nodes (SNs) that make up the WSN are all initially set to the same energy levels.
- For effective environmental monitoring, sensor nodes are positioned at random across the target scenario.
- After deployment, base station (BS) and sensor nodes stay motionless, guaranteeing a constant topology.
- The sensor nodes have non-replaceable and non-rechargeable power sources and are energy-constrained and unsupervised.
- The Received Signal Strength Indicator (RSSI) is applied to assess distance between nodes and the base station.

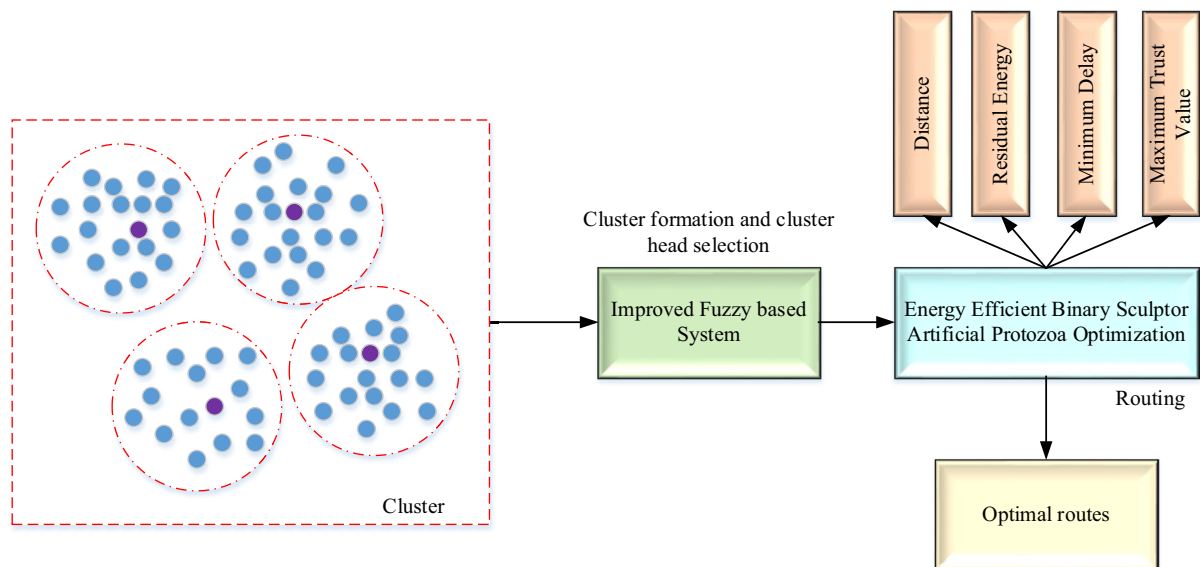


Fig. 1 Basic block diagram of the proposed model

- To replicate a real-world monitoring situation, base station is situated far from sensor deployment area.
- It is believed that base station's energy resources for communication and processing are limitless.
- Energy can be saved by each sensor node dynamically adjusting its transmission energy according to distance to intended receiver.
- When the energy cost of sending and receiving across distance is equal, radio communication channel is said to be symmetric.
- Energy depletion causes node failure and no maintenance is presumed after deployment.
- Since the network is built to create and send data continually, energy-efficient routing is necessary for extended operation.

3.2 Network Model

SNs are evenly and randomly distributed over a 100×100 m two-dimensional monitoring field in the suggested EE-BSAPO-based WSN architecture. The target field is separated into four areas according to the distance from the BS. Effective clustering and balanced energy use are supported by this geographic division. Energy-aware routing is essential for extending the network lifespan since base station is positioned far from sensing field. Because of longer communication distance, a region's nodes become more energy-constrained whichever distance it is from the base station. Cluster formation, CH selection and routing decisions are adjusted depending on nodes' proximity to the BS by dividing the deployment area into distance-based zones. The EE-BSAPO algorithm can dynamically choose the best routes by taking into account important criteria includes distance, residual energy, latency and trust values due to this geographical division, which also improves network's energy efficiency, scalability and routing efficacy.

3.3 Radio Model

The radio energy consumption model is utilized in this proposed EE-BSAPO-based routing protocol to assess energy consumption during data transmission and reception between sensor nodes. Nonetheless, Eqs. (1) and (2)

provide the basis of the behaviour. The amplifier energy in multipath, free space and electronics are represented by ε_{nq} , ε_{fh} and E_{elec} , respectively.

$$E_{T_y}(n, k) = \begin{cases} nE_{elec} + n\varepsilon_{fh}k^2 & k < k_0 \\ nE_{elec} + n\varepsilon_{nq}k^4 & k \geq k_0 \end{cases} \quad (1)$$

Where k denotes the separation distance between the transmitting and receiving node. $k_0 = \sqrt{\frac{\varepsilon_{fh}}{\varepsilon_{nq}}}$. If $k < k_0$ represents short distance communication and free space model is used. If $k \geq k_0$ represents long distance communication and multipath fading model is used.

The following energy is needed to receive an n -bit message:

$$E_{R_y}(n) = E_{R_{y-elec}}(n) = n \cdot E_{elec} \quad (2)$$

The network size $H = d * bsquare$ meters was considered in this study, and there were n SN dispersed over the specified region H . The BS is situated in a remote location, far from network H . Equation (3) may be used to calculate the energy used up by each CH for each round.

$$E_{Chead} = mn(E_{elec} + \varepsilon_{fh}k_{BS} + E_{AD}) \quad (3)$$

Where d is the separation between CH and BS. Likewise, Eq. (4) calculates the energy used up by each cluster member.

$$E_{CM} = n(E_{elec} + \varepsilon_{fh}k_{Chead}) \quad (4)$$

To measure mean distance between a CH and its members or k_{Chead} using Eq. (5). Each SN's transmission radius may be estimated using this mean distance.

$$k_{Chead} = \frac{1}{(m \setminus l)} \sum_J^{m \setminus l} k_j \quad (5)$$

Where l indicates the number of clusters, k_j denotes the separation between a cluster member J and CH and m represents the total number of SN. SNs are dispersed arbitrarily and indiscriminately throughout the target monitoring region in the suggested EE-BSAPO-based routing protocol. The protocol is triggered to start communication when the deployment is finished. Clusters are created at the initialization of each cycle to arrange sensor nodes for effective data collecting. The network operates in rounds. During each round, a maximum of $p\%$ of the SNs that are currently active (alive) are chosen as possible CHs. The BS transmits a control packet called LOC_BS across the network to start the procedure. In order to prevent collisions, this packet contains vital information including the location of the BS, the coordinates of the present areas (which are categorized according to their distance from the BS) and specific time slots for each SN to broadcast. Then, within its designated time slot, each SN broadcasts a Hello packet (*Hello_{packet}*). Each SN individually calculates local metrics like residual energy, node density and distance to BS once every node has finished broadcasting. The Improved Fuzzy-based System uses these characteristics as inputs to determine best cluster formation and CH selection. In order for the EE-BSAPO algorithm to carry out dependable and effective routing in later stages, this procedure guarantees a balanced and energy-conscious network organization.

3.4 Improved Fuzzy based System for clustering and cluster head selection phase

A FL method is incorporated into the suggested EE-BSAPO-based routing protocol to make it easier to select the best candidate nodes for CH position. When it comes to simulating human-like thinking and decision-making,

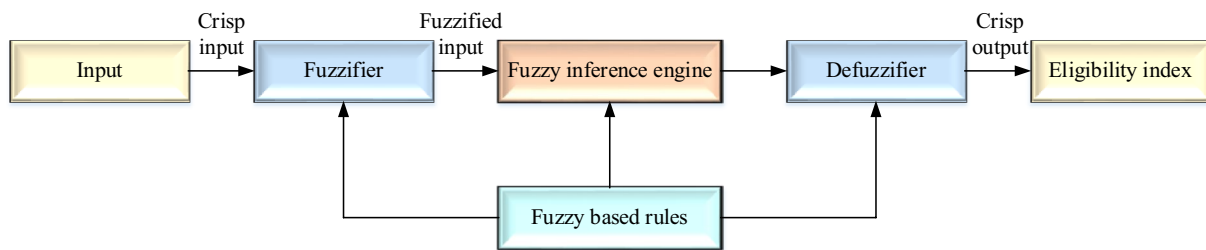


Fig. 2 Basic Block Diagram of Fuzzy System

Table 2 Input function of fuzzifier

Input	Variables		
Distance to base station	Close	Medium	Far
Residual energy	Low	Medium	High
Density of nodes	Sparse	Medium	Dense

fuzzy logic works incredibly well, particularly when there are ambiguity and several input factors. Figure 2 shows the four basic parts that make up the framework of the fuzzy system used in this work.

Fuzzifier This part creates fuzzy sets from sharp input values. Every input parameter is given a degree of membership in predetermined linguistic categories, including residual energy, node density and distance to base station. The system can now qualitatively evaluate input from the real world due to this modification.

Fuzzy Rule Base The system's operational logic is defined by a collection of IF-THEN rules. These guidelines define dynamic interactions between input variables and capture expert knowledge. An important factor knowledge base means directing decision-making in rule basis.

Inference Engine Inference engines process the fuzzy inputs to stimulate human thinking based on rules. Thus, it generates a fuzzy output that transfer system's conclusion after evaluate all rules and determining to which rule applies.

Defuzzifier It convert inference engine's fuzzy output into numerical number. Making decisions like choosing the best CH which requires this defuzzification process. While computing the defuzzified output using the centroid approach comprehensible and balanced outcome is attained.

In WSN, intelligent and adaptive CH selection is obtained by proposed protocol via combination of fuzzy logic system, which optimizes network lifespan and energy efficiency.

3.4.1 Fuzzification

Residual Energy, Node Density and Distance to BS are three eligibility criteria for sensor node to evaluate fuzzy-based cluster head selection mechanism. Fuzzification means converting these clear input values into matching fuzzy sets based on particular language concepts. Euclidean distance between sensor node and base station affect communication energy cost. Residual power shows capacity of sensor node to execute more communication tasks. The total number of nodes exists inside communication range is considered as CH nomination is referred to as node density it affects the cluster acceptability and node's centrality. Table 2 provides an overview of the lowest and maximum values of these parameters that were applied during fuzzification for membership function generation. The fuzzified inputs are utilized by fuzzy inference system to measure the eligibility criteria for selecting the best CHs.

3.4.2 Fuzzy Rule Base

The Fuzzy Rule Base means grouping of IF-THEN rules developed to evaluate sensor node's properness for Cluster Head (CH) selection, processes the membership values that emerge from fuzzifying the input variables. Residual Energy, Node Density, and Distance to BS are three input factors whose linguistic values are combined to create these rules. As shown in Table 3, a total of 27 rules are used, including every conceivable combination of input conditions.

The truth value of each rule is ascertained using fuzzy logic operators like fuzzy AND and fuzzy OR in order to extract an output from the rule base. Following the evaluation of the rules, the output fuzzy sets generated by each activated rule are combined using the aggregation approach. The largest value is chosen to reflect the combined inference result from this aggregated fuzzy output. The Mamdani inference system, which is frequently utilized in real-world applications because of its strong decision-making capabilities and easy-to-understand rule representation, is utilized in this proposed model for fuzzy reasoning. A fuzzy set is produced as the result of this fuzzy inference process and it is subsequently sent to the defuzzification step in order to calculate the final eligibility index. Sensor nodes for CH selection are ranked using this metric according to their local network density, closeness and energy efficiency.

3.4.3 Defuzzification

Table 4 shows the fuzzy variables that are utilized to produce the crisp value output.

According to Eq. (6), the centre of area (G^*) approach is applied for defuzzification.

Table 3 Fuzzy Rules

Rule	Residual energy	Distance to base station	Density of node	Eligibility index
1	Low	Far	Sparse	Low
2	Low	Far	Medium	Low
3	Low	Far	Dense	Low
4	Low	Medium	Sparse	Low
5	Low	Medium	Medium	Low
6	Low	Medium	Dense	Medium
7	Low	Close	Sparse	Low
8	Low	Close	Medium	Medium
9	Low	Close	Dense	Medium
10	Medium	Far	Sparse	Low
11	Medium	Far	Medium	Medium
12	Medium	Far	Dense	Medium
13	Medium	Medium	Sparse	Medium
14	Medium	Medium	Medium	Medium
15	Medium	Medium	Dense	High
16	Medium	Close	Sparse	Medium
17	Medium	Close	Medium	High
18	Medium	Close	Dense	High
19	High	Far	Sparse	Medium
20	High	Far	Medium	Medium
21	High	Far	Dense	High
22	High	Medium	Sparse	Medium
23	High	Medium	Medium	High
24	High	Medium	Dense	High
25	High	Close	Sparse	High
26	High	Close	Medium	High
27	High	Close	Dense	High

Table 4 output function of defuzzifier

Output	Variables
Eligibility index	Worst, bad, fair, good, better, far better, best, very best

Table 5 Specific parameters for all membership functions

(a) Gaussian Membership Functions					
Input Variable	Linguistic Term	Mean (μ)	Standard Deviation (σ)		
Residual Energy	Medium	0.5 E_o	0.15 E_o		
Distance to BS	Medium	50 m	15 m		
Node Density	Medium	10 nodes	3 nodes		
(b) Trapezoidal Membership Functions					
Input Variable	Linguistic Term	a	b	c	d
Residual Energy	Low	0	0	0.2 E_o	0.4 E_o
Residual Energy	High	0.6 E_o	0.8 E_o	E_o	E_o
Distance to BS	Close	0	0	20	40
Distance to BS	Far	60	80	100	100
Node Density	Sparse	0	0	5	8
Node Density	Dense	12	15	20	20

$$G^* = \frac{\int \eta_D(y) \cdot y \cdot dy}{\int \eta_D(y) dy} \quad (6)$$

Where $\eta_D(y)$ represents the output fuzzy set D 's aggregated membership function value was assessed at output variable y . Fuzzy systems frequently employ a variety of membership functions (MFs), include Gaussian, Sigmoid, Trapezoidal and Triangular functions. Any membership function's primary prerequisite is that its result must fall between 0 and 1. To improve the efficiency of fuzzy inference system, also, mix of Gaussian and Trapezoidal membership functions is applied. For intermediate values in particular, Gaussian membership functions are used because they offer seamless transitions and preserve non-zero values across the range, both of which are advantageous for correctly describing fuzzy sets. Trapezoidal membership functions, on the other hand, are used to represent boundary variables because of their structural simplicity and computing efficiency. The suggested EE-BSAPO-based routing protocol performed better in terms of classification accuracy and decision reliability when using a mix of Gaussian and Trapezoidal membership functions, while other membership functions may also be used. Equations (7) and (8) provide the Gaussian and trapezoidal MF that are included in our FIS, respectively.

$$\eta_{gaus}(y) = e^{\frac{-(y-s)^2}{2\sigma^2}} \quad (7)$$

$$f(y; a, b, c, d) = \max\left(\min\left(\frac{d-y}{d-c}, 1, \frac{y-a}{b-a}\right), 0\right) \quad (8)$$

The trapezoid's feet and shoulders are denoted as d , a and c , d , respectively. The eligibility index for every node is determined when defuzzifier transforms supplied input into crisp set. Table 5 represents the specific parameters for all membership functions.

A dynamic threshold adjustment method based on network's operating state is benefit to enhance the fuzzy-based CH selection system. In this model, CH selection is evaluated by static or mean-based eligibility threshold (TH). Nevertheless, this model ignores the network's aging and fluctuating energy conditions. Frequent explosion of nodes with lower residual energy from CH selection then it moves through many rounds because of its uneven energy consumption and early node failures. A dynamic threshold ($T_{dynamic}$), adjustment is based on every node

residual energy and current round number. It is obtained by grouping mean eligibility index with a scaling factor to determine network's lifespan and node's remaining energy. The dynamic threshold is defined in Eq. (9):

$$T_{dynamic} = \delta \cdot mean(EJ) + (1 - \delta) \cdot \left(\frac{Re_j}{Re_{max}} \cdot \left(1 - \frac{r}{r_{max}} \right) \right) \quad (9)$$

Where r indicates the current round, r_{max} indicates the total number of rounds, Re_j provides the residual energy of node j , Re_{max} denotes greatest residual energy among all nodes and δ represents a weighting factor (e.g., 0.6). High-energy nodes are more likely to be chosen as CHs in early rounds because to this dynamic threshold, while later rounds allow even moderately energetic nodes to contribute as CHs because of a loosened threshold. In the end, this adaptation increases network longevity and stability by improving load distribution and lowering the probability of early node mortality.

The CH candidates announce their selection by broadcasting a message with the label CH_{Head} throughout their communication range (R_c) once they have been identified during the topology-building phase. In response, the remaining nodes that are not chosen to be CHs associate themselves with the most suitable cluster by sending a CM_{join} message to the closest CH. Consequently, a collection of popt clusters is constantly generated according to network circumstances and closeness. After receiving an acknowledgment from a neighbouring CH, nodes that are still unassociated after the first cluster creation join the nearest cluster that becomes accessible. Once all sensor nodes have been successfully allocated to clusters, the topology configuration process is deemed complete, guaranteeing an organized and energy-efficient network topology.

In conclusion, the suggested EE-BSAPO-based WSN protocol's energy efficiency and overall performance are greatly improved by the clustering and CH selection process employing the Improved Fuzzy-based System. A more smart and perceptive CH selection is made possible by the fuzzy logic system's incorporation of numerous choice characteristics, including node density, residual energy, and distance to base station. The Mamdani inference technique effectively processes the fuzzy rule base to give a robust eligibility index, also, the use of Gaussian and Trapezoidal membership functions guarantees smooth and computationally efficient fuzzification. In addition to select CHs, this flexible and dynamic strategy also helps to balance energy usage, lower communication overhead, and extend network lifetime. Fuzzy-based clustering method provide a strong foundation for the subsequent EE-BSAPO routing phase for *scalable and reliable data transfer inside the WSN*.

3.5 Energy Efficient Binary Sculptor Artificial Protozoa Optimization for Routing Phase

Efficient routing is important for WSNs in to minimize power consumption, lower communication overhead and improve network durability. The proposed routing architecture create the best communication channels between CHs and BS after cluster selection stage. An Energy Efficient Binary Sculptor Artificial Protozoa Optimization (EE-BSAPO) method is used into the proposed work to solve issues of dynamic topology, limited energy and reliable data transfer. To select the best route multi objective fitness function which depends on factors such as distance, latency, trust value and residual energy. Based on these factors, the routing model maintains connection and increase efficiency of data transmission throughout.

3.5.1 Fitness Function

In EE-BSAPO-based protocol, the best routing path is detected by analysing a composite fitness function based on factors, including distance, trust value, residual energy and end-to-end latency. These parameters validation are used for selection of CH for transmission of data from source to destination. The path has highest fitness value is considered as best node during data transmission. Greater residual energy and trust reduce distance and communication latency and improve fitness value. For EE-BSAPO algorithm select reliable, low-latency and energy-efficient routes, the maximum fitness function is developed. Equation (10) is represented as follows:

$$F = \frac{1}{4} [f + (1 - k) + (1 - \alpha) + \eta] \quad (10)$$

$$f = \frac{1}{c} \sum_{L=1}^c f_L \quad (11)$$

$$k = \frac{c}{n} \quad (12)$$

$$\alpha = \frac{1}{c^2 \times \varphi} \sum_{L=1}^c \sum_{\substack{S=1 \\ S=L+1}}^c {}^c\alpha(L, S) \quad (13)$$

$$\chi = \frac{1}{3c^2 \times \varphi} \sum_{L=1}^c \sum_{\substack{S=1 \\ S=L+1}}^c [RT + DT + HT] \quad (14)$$

Where χ stands for trust, ϕ for normalization factor, f for energy, k for delay, α for distance and F for fitness function. Equation (14) in the EE-BSAPO framework models trust as a dynamic and quantifiable characteristic that directly affects the composite fitness function. The trust value χ is calculated by aggregating three behavioural indicators: recent trust (RT), direct trust (DT), and historical trust (HT) across all nodes in the cluster. This is normalized mathematically using the factor $3c^2 \times \phi$ to guarantee that χ stays confined inside $[0, 1]$. Every element of trust is updated iteratively: direct trust shows the success percentage of interactions, historical trust incorporates longer-term stability, and current trust represents the most recent forwarding behaviour. In order to keep stale behaviour from taking over the evaluation, these trust components naturally deteriorate over time. This decay is managed via exponential attenuation, which causes older trust values to decrease at a rate proportionate to their age, guaranteeing that the system gives priority to current network behaviour. When these components are combined, the trust metric becomes resilient and responsive, allowing the fitness function F to favour cluster heads and routes that exhibit minimal packet loss or malicious activity risk and constant reliability. By including trust into the fitness model, EE-BSAPO is able to eliminate unreliable nodes and adaptively favour trustworthy ones, improving network performance and routing stability.

The Euglena protozoan behaviour such as heterotrophic foraging, autotrophic foraging, reproduction and dormancy are used as the model for the BSAPO algorithm. By using these behavioural patterns the exploration and exploitation phase are successfully balanced during optimization process. In specific, honing local optima (exploitation) is focused by reproduction and heterotrophic, also global search capabilities are optimized by autotrophic foraging and dormancy.

The continuous search space in existing Artificial Protozoa Optimization (APO) works its direct application to discrete or binary optimization issues, including cluster-based routing path selection in WSNs. The proposed work presents a Binary version of the APO algorithm (BSAPO) developed for discrete optimization to address issues. Search agent present in this environment is described as a binary vector with string of 0s and 1s. Every bit in this vector encodes a decision like whether a node should be included in the best path or not. V-shaped and S-shaped transfer functions are utilized to modify continuous position update mechanisms of APO algorithm to increase efficient binary optimization. Every search agent may efficiently travel to the binary solution space which covert continuous search values to binary values. To detect most dependable and energy-efficient routing routes, BSAPO warrantee both variety in exploration and precision in exploitation.

Each search agent is assessed using a fitness function that takes residual energy, distance, latency and trust value into account. The initial population of binary solutions is created at random. By iteratively updating the population, the method progressively converges to an optimum or nearly optimal routing configuration while maintaining the best solutions.

3.5.2 Foraging Autotrophically

The mechanism by which euglena generate nutrients using chloroplasts is known as autotrophic foraging. In low light, the creature exhibits phototropism, moving toward higher light intensity, but in bright light, it prefers lower light intensity, displaying photophobia. From Eq. (15) to (18) represents the mathematical equation for autotrophic foraging:

$$Y_j^{new} = Y_j + f \cdot Y_i - Y_j + \frac{1}{np} \cdot \sum_{l=1}^{np} W_p \cdot Y_{l-} - Y_{l+} \Theta N'_f \quad (15)$$

$$f = rand \cdot 1 + \cos \frac{iter}{iter_{max}} \cdot \pi \quad (16)$$

$$W_p = e^{-\frac{fY_{l-}}{fY_{l+} + eps}} \quad (17)$$

$$N_f dj = \begin{cases} 1, & \text{if } dj \text{ is in rand perm} \left(\dim, \left[\dim \cdot \frac{j}{ps} \right] \right) \\ 0, & \text{otherwise} \end{cases} \quad (18)$$

Equation (15) shows that pairwise neighbor interactions, controlled by the autotrophic foraging coefficient W_p , dictate location updates in autotrophic foraging. Mapping vector in foraging is provided as in N_f and the dimension index is defined as dj . If the neighbor pair's ranking index is $i < j$, it indicates that the j^{th} euglena is at a lower light intensity and is phototropic; if the neighbor pair's ranking index is $i > j$, it indicates that the j^{th} euglena is at a higher light intensity and is photophobic. To enhance the optimization model, Sculptor Optimization Algorithm (SOA) is introduced. The optimal shape is represented by the sculpting pattern, which is the population's best option according to the SOA. Each member of the population moves in the direction of this ideal pattern, adjusting its location in the search space, much how a sculptor refines a raw material to meet a desired design. This modification simulates the sculpting procedure, which involves making small adjustments to enhance alignment with the objective. Equation (19) simulates this transition by computing a new location for each solution in the search space. Equation (20) states that the solution is updated to the improved location if the newly calculated position produces a greater objective function value than the one that was obtained previously. The population converges toward the ideal solution after this repeated refinement, which successfully strikes a balance between exploration phase and exploitation phase in the search space.

$$Y_{j,i}^{new} = y_{j,i} + r \cdot (best_i - Q \cdot y_{j,i}) \quad (19)$$

$$Y_j = \begin{cases} Y_j^{new}, & F_j^{new} \leq F_j \\ Y_j, & \text{else} \end{cases} \quad (20)$$

Where $best$ represents the population's best member, $best_i$ constitutes its i^{th} dimension, Y_j^{new} represents the j^{th} member's new position based on the first phase of SOA, $Y_{j,i}^{new}$ represents its i^{th} dimension, F_j^{new} is its objective function value, r constitutes a randomly selected number from the interval $[0, 1]$ and Q represents a randomly chosen number, taking values of 1 or 2.

3.5.3 Foraging by Heterotrophs

Euglena use a technique known as heterotrophic foraging to get nutrients in dark conditions by consuming organic materials. The creature goes in the direction of Y_j^{new} , supposing that there is a nutrient-rich area nearby. Random vector with elements in $[0,1]$ is represents as $Rand$. The position update can be defines in Eq. (21) to (24):

$$Y_j^{new} = Y_j + f \cdot Y_{near} - Y_j + \frac{1}{np} \cdot \sum_{l=1}^{np} W_u \cdot Y_{j-l} - Y_{j+l} \Theta N_f \quad (21)$$

$$Y_{near} = 1 \pm rand \cdot 1 - \frac{iter}{iter_{max}} \Theta Y_j \quad (22)$$

$$W_u = e^{-\frac{fY_{j-l}}{fY_{j+l}+eps}} \quad (23)$$

$$Rand = rand_1, rand_2, \dots, rand_{dim} \quad (24)$$

3.5.4 The state of Dormancy

Euglena go into hibernation as a survival tactic when they are subjected to environmental stress. In order to pre-serve population variety and prevent local optima, this procedure entails replacing dormant euglena with freshly created individuals. The following Eqs. (25) and (26) is the mathematical model of dormancy:

$$Y_j^{new} = Y_{min} + Rand \Theta Y_{max} - Y_{min} \quad (25)$$

$$Y_{min} = lb_1, lb_2, \dots, lb_{dim} \quad Y_{max} = ub_1, ub_2, \dots, ub_{dim} \quad (26)$$

The APO algorithm is qualified to leap out of local optima due to the random formation of domains that would be determined by the production of new euglena during dormancy.

3.5.5 Reproduction

Asexual reproduction occurs when euglena achieve their ideal age and health. This method is implemented in our simulation by creating a duplicate euglena with controlled perturbation, even though the process theoretically yields two identical offspring. The following Eqs. (27) and (28) is the mathematical expression for this reproductive process:

$$Y_j^{new} = Y_j \pm rand \cdot Y_{min} + rand \Theta Y_{max} - Y_{max} \Theta N_r \quad (27)$$

$$N_r dj = \begin{cases} 1, & \text{if } dj \text{ is in } rand \text{ perm}(\dim, [\dim \cdot rand]) \\ 0, & \text{otherwise} \end{cases} \quad (28)$$

Where Mapping vector in reproduction is represented as N_r . The necessity to overcome two practical drawbacks that APO has when used in actual WSN routing circumstances is what led to the switch from the traditional APO algorithm to the suggested BSAPO. While APO works well in continuous search spaces, it lacks the structural accuracy needed for binary decision-making processes like node-state selection, route activation and cluster-head election. Current binary extensions of APO do not significantly alter the behavioral dynamics of protozoa for energy-sensitive networks; instead, they just discretize the search space. BSAPO presents a binary movement mechanism inspired by sculptors that improves convergence in high-dimensional binary search areas, minimizes redundant transitions and compresses exploration pathways. Instead of depending solely on probabilistic flipping,

this binary sculpting allows biological agents to make more meaningful and precise choices. As a result, BSAPO is more suited for WSN routing, where choices are essentially binary (e.g., trust/not trust a node, activate/deactivate a connection, or pick/not choose a cluster head). Thus, this smooth transition from APO to BSAPO offers both a context-aware improvement suited to the practical requirements of energy-constrained sensor networks as well as a mathematically coherent progression. The Pseudocode for proposed routing models is given as:

Algorithm 1: Pseudocode for the proposed routing model

Input: M = Sensor Node Count, BS Base Station position, $Max_{iter} \leftarrow$ Maximum iteration count, D =Dimensionality (as a function of the number of CHs), fitness parameters
Output: Optimal Routes: A collection of the best routes from CHs to BS
Initialize: $Z \rightarrow$ population size
Set up the binary population $Y[j][d]$ initially
For $j = 1$ to P and $d = 1$ to D
 Assess each population solution's fitness using equation (10)
 Update the best solution with highest fitness
 $1 \rightarrow iter$
 While $iter \leq Max_{iter}$ **do**
 For every search agent j in population **do**
 Phase of Autotrophic Foraging
 Update the position using equation (15) to (18)
 Apply equation (19) and (20) for improving the foraging stage
 To convert to binary, use an S-shaped or V-shaped transfer function to Y_j^{new} .
 Foraging in Heterotrophic Conditions
 Position should be updated toward the nutrient-rich area using
Equation (21) to (24).
 Use binary transfer and calculate
 The Phase of Dormancy
 If Dormant Condition Met **then**
 Using Equation (25 to 26) to substitute a newly created random
binary vector for $Y[j]$.
 End if
 Reproduction Stage
 If $Age[j] \geq threshold\ age$ and $Fitness[j] \geq reproduction\ threshold$
 Modify $Y[j]$ slightly to create a new solution
 End If
 Update optimal solution if better fitness is establish
 $1 + iter \rightarrow iter$
 End For
 End While
To obtain Optimal Routes, decode the optimal binary vector
For every byte in optimal solution
 If $bit = 1$ **then**
 Add the appropriate CH-to-BS path to Optimal Routes
 End If
End For
Return Optimal Routes
End

While traditional APO, Binary-APO and Sculptor-based optimization algorithms offer robust global search capabilities, they frequently find it difficult to adjust to the quickly evolving energy and trust dynamics of real-world WSNs. Similar to this, hybrid fuzzy–metaheuristic clustering techniques enhance local decision-making

Table 6 Hyper-parameters

Hyper-parameters	
Transmission range	25–50 m
Initial node energy	2 J
Number of nodes	1000
Deployment area	100 × 100
Initial trust value	1.0
Mutation rate	0.01–0.05
Number of iterations	100–500

Table 7 Adversarial model for simulation

Parameter	Value
Percentage of malicious nodes	10% – 30% of total nodes
Type of malicious behaviour	Selective packet dropping
Attack model	Internal (compromised sensor nodes)
Packet forwarding behaviour	Random and selective dropping
Trust degradation basis	Packet loss rate, forwarding failure, abnormal delay
Trust update method	Dynamic, round-based
Impact on routing	Low-trust nodes excluded from CH and routing paths

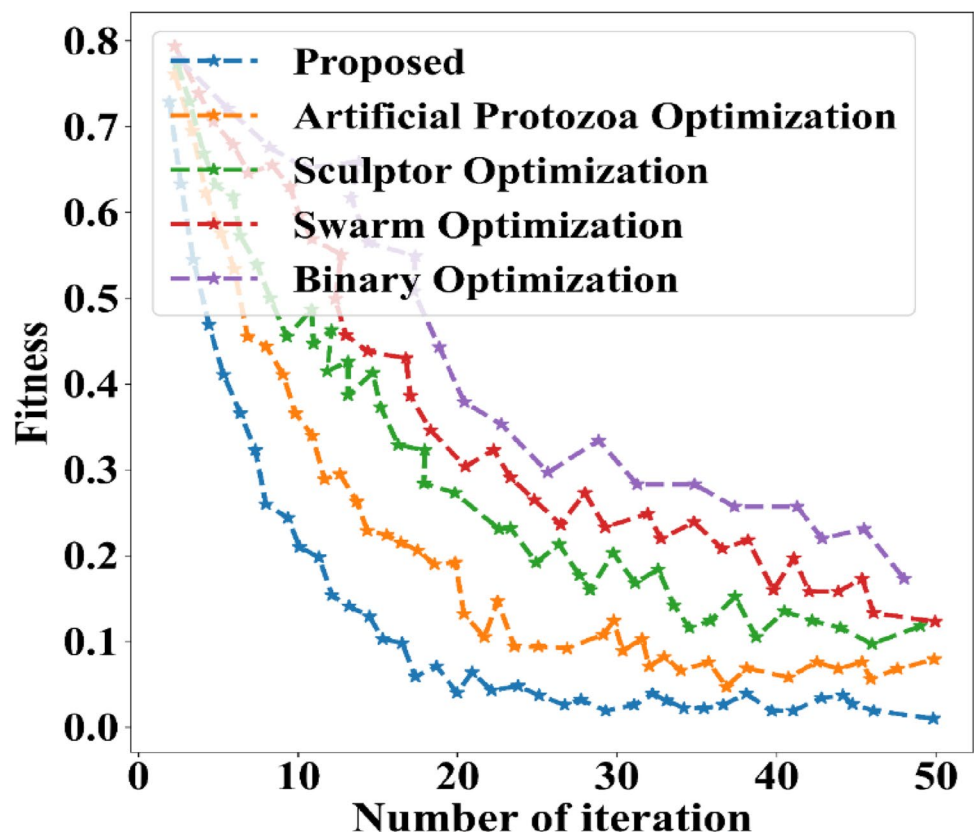
but continue to rely on static rules or short optimization cycles, which limit their responsiveness and eventually deteriorate routing stability. The proposed EE-BSAPO is new because it combines a Binary Sculptor Artificial Protozoa Optimization approach with an enhanced fuzzy inference system in a synergistic way to create a single framework that makes nodes more self-adaptive and context-aware. EE-BSAPO incorporates fuzzy-weighted intelligence directly into both cluster-head selection and route evolution, in contrast to previous APO or SOA variations. This enables the optimizer to react dynamically to multi-criteria inputs like residual energy, communication latency, trust fluctuations and inter-node distance. This leads to a routing system that is more durable, energy-balanced and behaviour-aware than existing methods as it not only searches effectively but also comprehends the network's current state. The suggested EE-BSAPO-based routing architecture shows a strong and flexible way to provide dependable and energy-efficient communication in WSN. The model successfully strikes a balance between exploration phase and exploitation phase during route optimization by utilizing the sculptor's refinement mechanism and the hybrid behaviours of the Euglena protozoan. Its transfer functions and binary encoding enable accurate decision-making in the discrete routing space, guaranteeing the best possible path selection. The routing protocol may dynamically adjust to network circumstances, minimize packet loss and increase the total network lifetime by utilizing numerous fitness characteristics, including residual energy, latency, distance and trust value. This technique offers high packet delivery speeds, robust connection under dynamic topologies and a considerable reduction in energy usage as compared to traditional methods.

4 Results and Discussion

The proposed scheme is executed in the NS-3 tool with the system specifications of The 8 GB RAM with windows 11 using i5-4670 CPU @ 3.40 GHz which a system type of 64 bit operating system × 64 based processor. The system have no pen and touch input for the display. Performance of proposed method is evaluated and compared with previous models based on some performance metrics like delay, residual energy, throughput, trust, total energy consumption, network lifetime etc. Table 6 represents the hyper-parameters of the proposed model. Table 7 represents the adversarial model for simulation. Table 8 represents the simulation parameters of the proposed model.

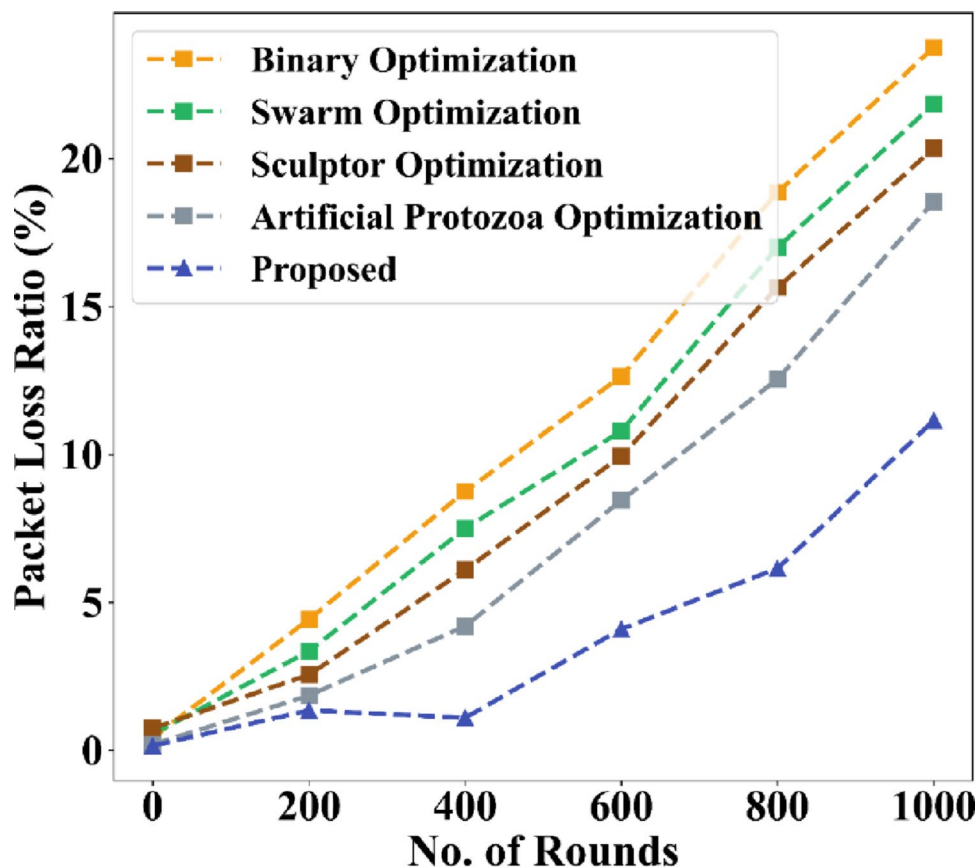
Table 8 Simulation parameters

Simulation Parameter	Value
Simulator	NS-3
Network area	100 × 100 m ²
Number of nodes	100–1000
Initial energy per node	2 J
Radio energy model	First-order radio model
Packet size	Same for all protocols
Traffic type	Constant Bit Rate (CBR)
Base station position	Fixed, outside sensing field
Trust evaluation model	Identical for all protocols
Malicious node ratio	Same across all protocols
Simulation rounds	1000
Performance metrics	Delay, Energy, Throughput, Trust, Lifetime

Fig. 3 (a) Convergence graph of the proposed and existing model, (b) Convergence graph of the proposed model with various binary optimisation models


4.1 Performance Evaluation using Various Models

The convergence behaviour of the suggested model over 50 iterations is shown in Fig. 3 (a) however, various optimization techniques such as Artificial Protozoa Optimization (APO), Sculptor Optimization, Swarm Optimization and Binary Optimization. In order to minimize the routing cost in WSNs, the fitness value is a measure of the quality of the solution; a lower value implies better convergence toward the ideal solution. While other methods plateau at higher levels, the suggested method shows the fastest and most steady convergence across all studied models, achieving a fitness value of around 0.05 by the 25th iteration. This quick decline in fitness demonstrates the enhanced optimization power of the suggested EE-BSAPO model, which combines the advantages of the Sculptor and Artificial Protozoa Optimization processes in a binary framework. The outcome validates the model's effectiveness in detecting reliable and energy-efficient routing routes in a timely manner, minimizing

Fig. 4 PLR of the proposed and existing models**Table 9** PLR comparison of various models

Packet Loss Ratio						
Techniques	0	200	400	600	800	1000
Binary	0.43	4.43	8.75	12.65	18.87	23.76
Swarm	0.55	3.35	7.5	10.8	17	21.85
Sculptor	0.75	2.55	6.1	9.95	15.65	20.35
Artificial	0.2	1.85	4.2	8.45	12.55	18.55
Proposed	0.15	1.35	1.1	4.1	6.15	11.15

needless calculations, and preventing premature convergence. In comparison to conventional metaheuristic algorithms, the suggested method for optimum routing in WSNs turns out to be more efficient and dependable. Figure 3 (b) represents the Convergence graph of the proposed model with various binary optimisation models. The models such as Binary GWO, Binary PSO and Binary GO are used to compare with the proposed models. The proposed model attained better convergence value than existing models.

Comparing EE-BSAPO proposed model to other optimization techniques as Binary, Swarm, Sculptor, and Artificial Protozoa Optimization, the study of the Packet Loss Ratio (PLR) over several rounds shows how well it performs in Fig. 4. With just 11.15% at 1000 rounds, the suggested model continuously maintains the lowest PLR when the number of rounds rises from 0 to 1000. In contrast, Binary, Swarm, Sculptor, and Artificial Protozoa Optimization achieve higher values of 23.76%, 21.85%, 20.35% and 18.55%, respectively. The effective cluster head selection and trust-aware routing approach of the suggested architecture, which together reduce transmission failures and provide dependable data delivery, are responsible for this significant decrease in packet loss. Table 9 represents the comparison of PLR of various models.

Fig. 5 Trust comparison of various models

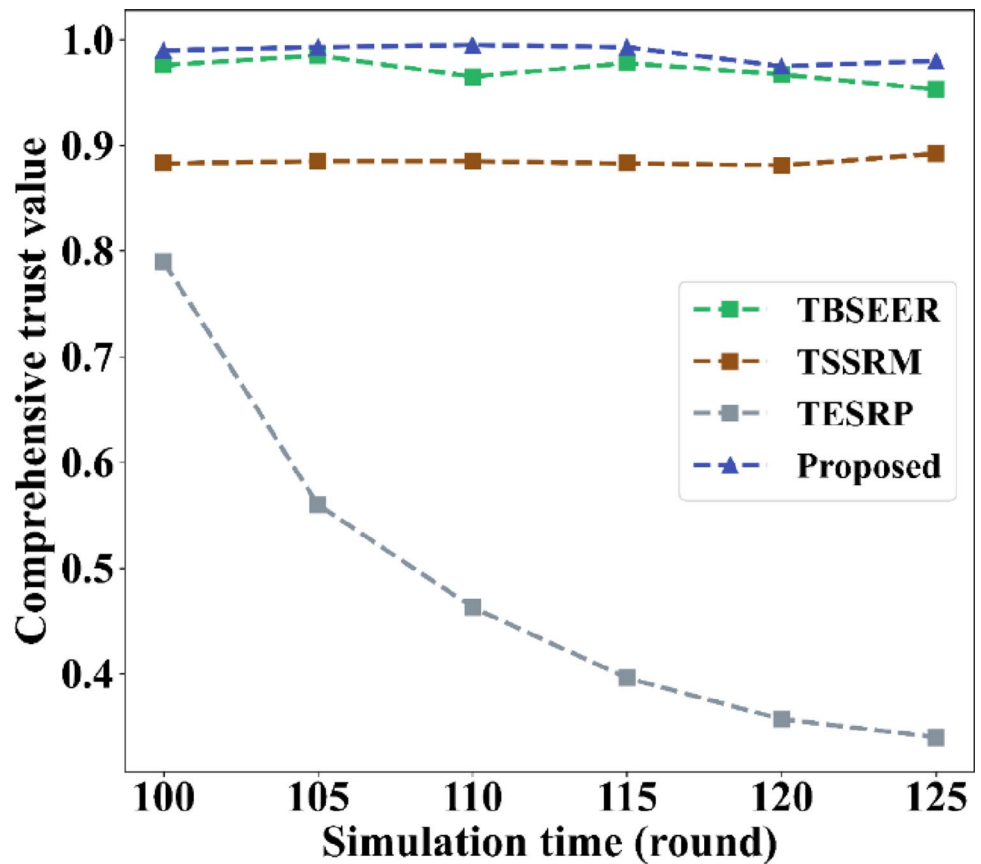


Table 10 The comparison of trust of various models

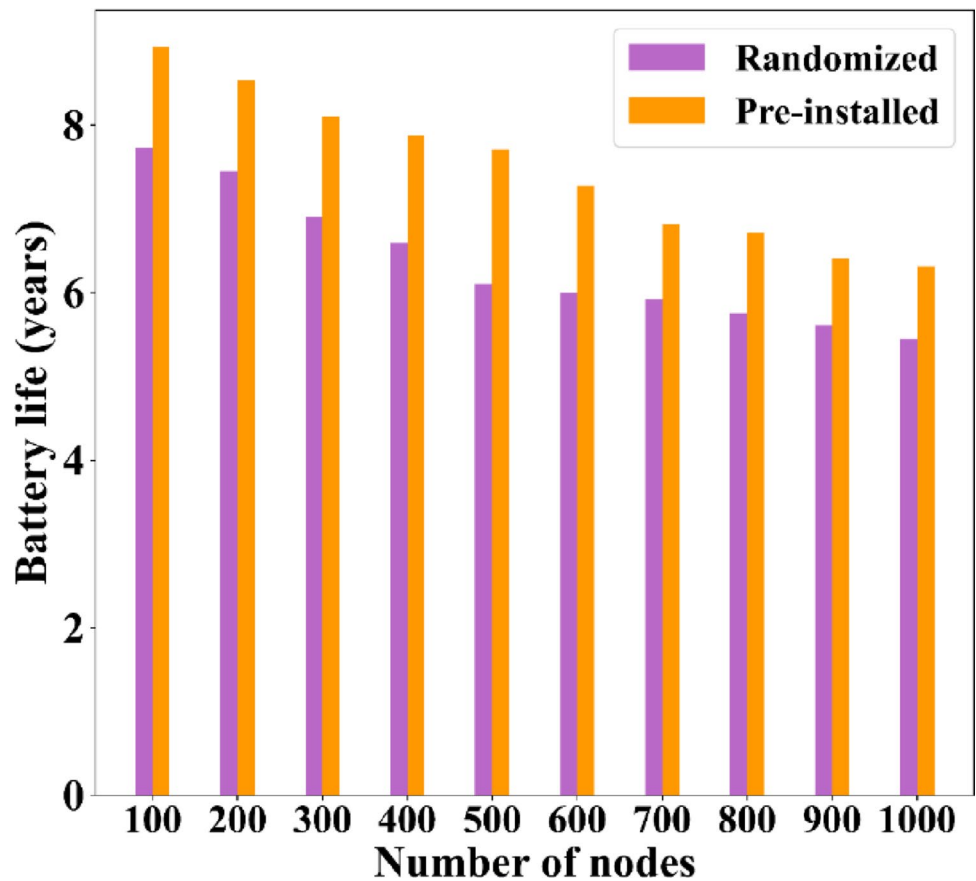
Comprehensive Trust value						
Technique	100	105	110	115	120	125
TBSEER	0.976	0.985	0.965	0.978	0.967	0.953
TSSRM	0.883	0.885	0.885	0.883	0.881	0.892
TESRP	0.79	0.56	0.463	0.396	0.357	0.34
Proposed	0.99	0.993	0.995	0.993	0.975	0.98

Overall trust value analysis across course of the simulation rounds given in Fig. 5 shows the way the proposed routing architecture performs in preserving high levels of trust throughout the network. The suggested model continuously records the greatest trust values, beginning at 0.99 in round 100 and staying above 0.97 throughout, culminating at 0.98 in round 125, as seen in the figure and accompanying data. This performance is far better than TESRP, which drops sharply from 0.79 to 0.34, and much better than TBSEER and TSSRM, which vary between 0.88 and 0.98. The suggested approach's resilience in safe data transfer is demonstrated by its capacity to maintain trust levels with little fluctuation, particularly throughout extended simulation times. This suggests that the suggested model's trust-aware routing approach successfully reduces hostile activity and encourages dependable communication, improving the WSNs overall credibility. Table 10 represents the comparison of trust of various models.

Across all node densities between 100 and 1000, the pre-installed node configuration consistently performs better than the randomized deployment method, according to the battery life evaluation shown in the chart and data. In particular, the pre-installed method's battery life begins at 8.929 years (for 100 nodes) and progressively drops to 6.319 years (for 1000 nodes), whereas the randomized strategy's lifespan is significantly less, falling from 7.735 to 5.445 years over the same range. This continuous benefit emphasizes how pre-installation allows

Table 11 Battery life comparison in years

Randomized	100	200	300	400	500	600	700	800	900	1000
	7.735	7.451	6.907	6.594	6.107	5.998	5.918	5.751	5.612	5.445
Pre-installed	8.929	8.529	8.101	7.875	7.708	7.279	6.821	6.712	6.399	6.319

Fig. 6 Battery life of proposed model

for improved routing stability and more organized energy utilization, which reduces overhead and retransmissions. As a result, the suggested pre-installed configuration guarantees a long operating lifetime in WSNs, which makes it ideal for extensive, energy-sensitive installations. Table 11; Fig. 6 represents the battery life comparison of randomized and pre-installed stage.

4.2 Performance Evaluation by Varying Network Size

In all network sizes (50 to 150 nodes) shown in Fig. 7, the suggested model performs noticeably better than all current routing protocols, including LEACH, CCBPR, PEGASIS, NCBPR, HHCA, ETLHCM and MLRP-HTEEN-UDSP [32], according to the energy efficiency analysis. Even while all models demonstrate an increase in power efficiency as network size grows, suggested model consistently and clearly leads. Its initial energy efficiency, at a network size of 50, is 49.799 J, which is already greater than the best of the rest. As the network grows, the efficiency increases gradually and reaches a high of 90.905 J for 150 nodes, exceeding the subsequent best model, MLRP-HTEEN-UDSP, which only achieves 86.378 J. This distinct difference across all sizes shows that the suggested approach not only scales better but also makes more efficient use of energy as node density rises. It reduced energy dissipation during transmission, effective cluster management, and optimal routing choices are all responsible for its exceptional performance. Thus, the suggested strategy turns out to be the

Fig. 7 Energy efficiency of the proposed and existing models

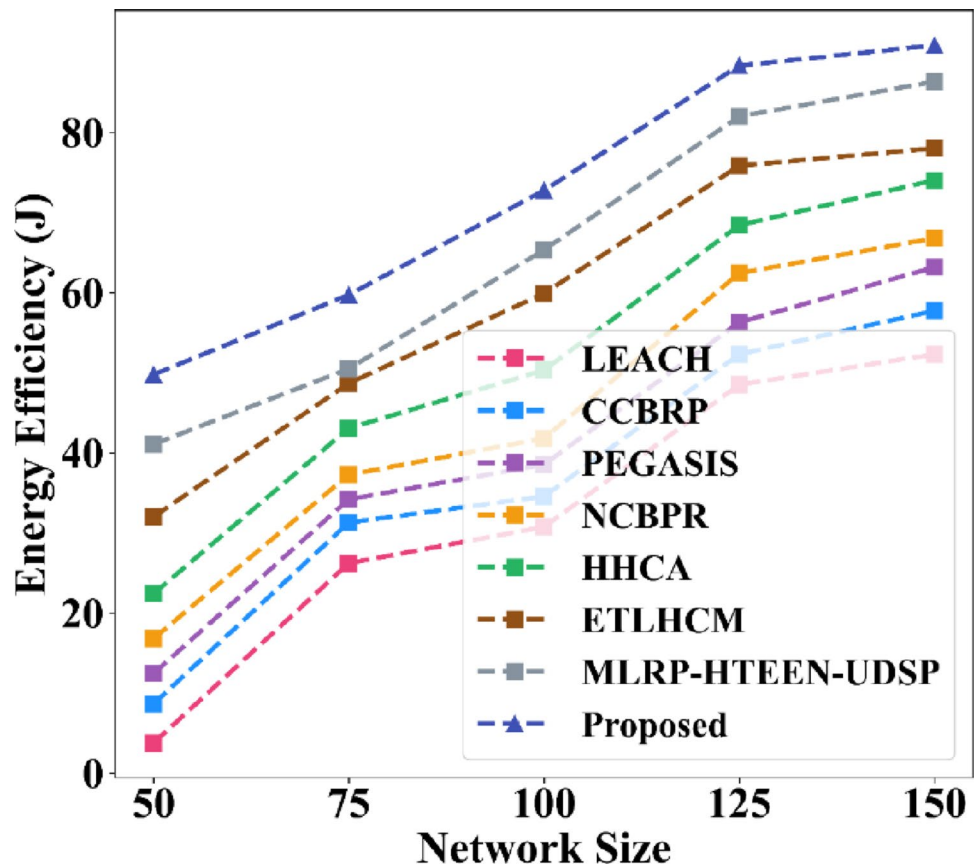
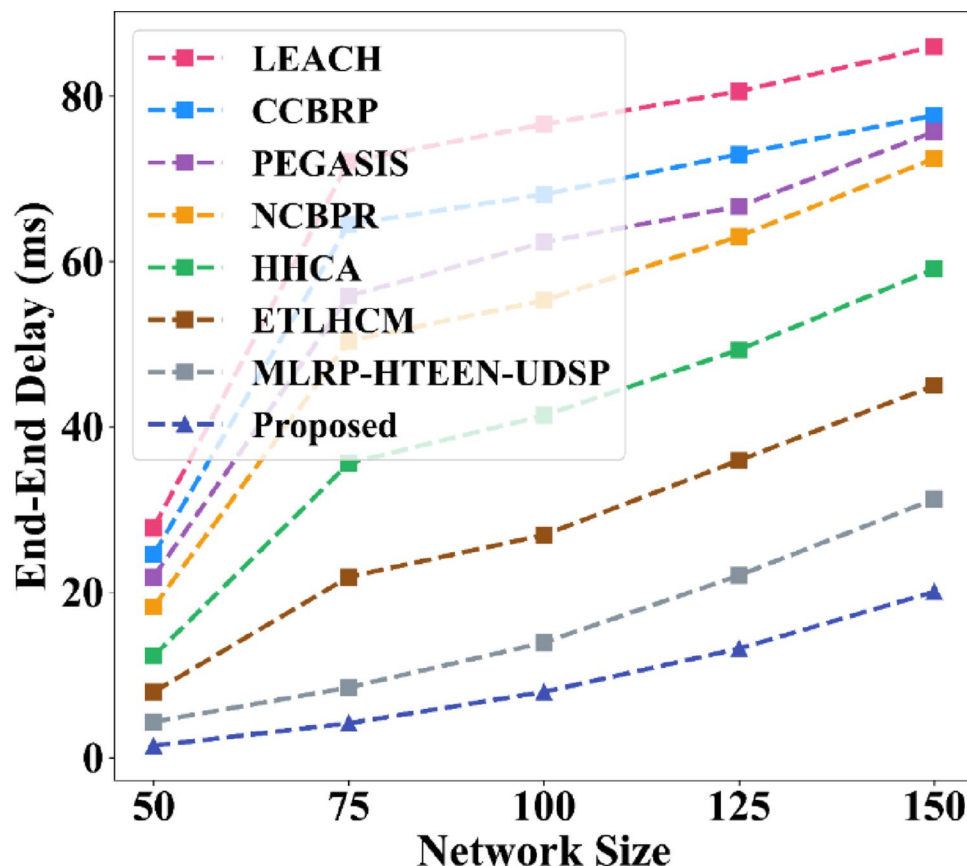


Table 12 Energy efficiency of various models

Energy Efficiency (J)					
Techniques	50	75	100	125	150
LEACH	3.803	26.258	30.785	48.531	52.334
CCBRP	8.692	31.328	34.588	52.334	57.767
PEGASIS	12.495	34.225	38.571	56.318	63.199
NCBPR	16.841	37.304	41.831	62.475	66.821
HHCA	22.455	43.099	50.342	68.451	74.064
ETLHCM	32.052	48.712	59.94	75.875	78.048
MLRP-HTEEN-UDSP	41.107	50.523	65.372	82.032	86.378
Proposed	49.799	59.759	72.797	88.37	90.905

best way to achieve long-term and scalable energy efficiency in WSN systems with energy constraints. Table 12 represents the comparison of energy efficiency (J) among various models.

Across a range of network sizes, proposed technique greatly reduces end-to-end latency is analysed to current techniques in Fig. 8. The suggested approach obtains an exceptionally low latency of 1.446 ms at a network size of 50 nodes, whereas baseline LEACH protocol achieves a high 27.831 ms, MLRP-HTEEN-UDSP 4.337 ms and ETLHCM 7.952 ms. The suggested technique maintains low latency with a delay of only 20.06 ms as the network grows to 150 nodes. This is significantly less than that of MLRP-HTEEN-UDSP (31.265 ms), ETLHCM (45.0 ms), HHCA (59.096 ms) and conventional routing protocols like PEGASIS (75.723 ms) and LEACH (86.024 ms). This steady performance indicates that the suggested approach is reliable and scalable in lowering transmission latency, is important for time-sensitive WSN applications. The main causes of delay decrease are clever cluster building techniques and improved routing routes, which lower transmission overhead and congestion. Table 13 represents end to end delay of proposed and existing models.

Fig. 8 End to end delay of the proposed and existing models**Table 13** End to end delay of the proposed and existing models

End-End Delay					
Techniques	50	75	100	125	150
LEACH	27.831	72.108	76.627	80.602	86.024
CCBRP	24.578	64.518	68.133	73.012	77.711
PEGASIS	21.867	55.843	62.349	66.687	75.723
NCBPR	18.253	50.422	55.301	63.072	72.47
HHCA	12.289	35.602	41.386	49.337	59.096
ETLHCM	7.952	21.867	26.928	35.964	45
MLRP-HTEEN-UDSP	4.337	8.494	13.916	22.048	31.265
Proposed	1.446	4.157	7.952	13.193	20.06

The proposed model results current routing methods in terms of network lifetime across all evaluated network sizes, as shown in Fig. 9. A lifespan of 49.799 units is attained by the suggested model at a network size of 50 nodes, which is substantially higher than that of MLRP-HTEEN-UDSP (40.563), ETLHCM (34.044), HHCA (27.344) and LEACH (2.177). As the network grows to 150 nodes, the suggested technique retains its superiority, with a peak lifespan of 88.913, considerably outperforming MLRP-HTEEN-UDSP at 85.473, ETLHCM at 80.765 and NCBPR at 64.467. The energy-aware routing approach and effective cluster head selection process in EE-BSAPO are responsible for this noteworthy improvement. Together, they balance the energy load and cut down on redundant transmissions. The concept is appropriate for large-scale, energy-constrained WSN setups where sustaining network activity for extended periods of time is crucial since it may increase operational length without reducing performance. Table 14 represents network life time of proposed and existing models.

Figure 10 represents the throughput of the proposed and existing models by varying network size which the suggested model's throughput performance far outperforms that of current methods, demonstrating improved

Fig. 9 Network life time of the proposed and existing models

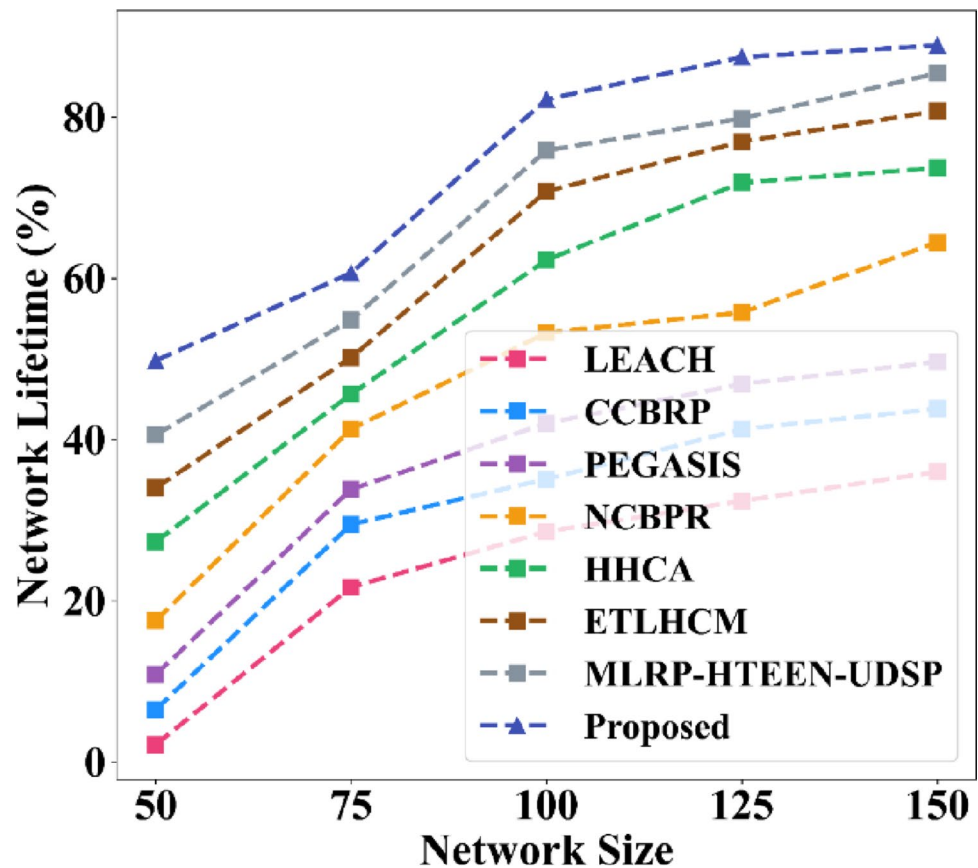
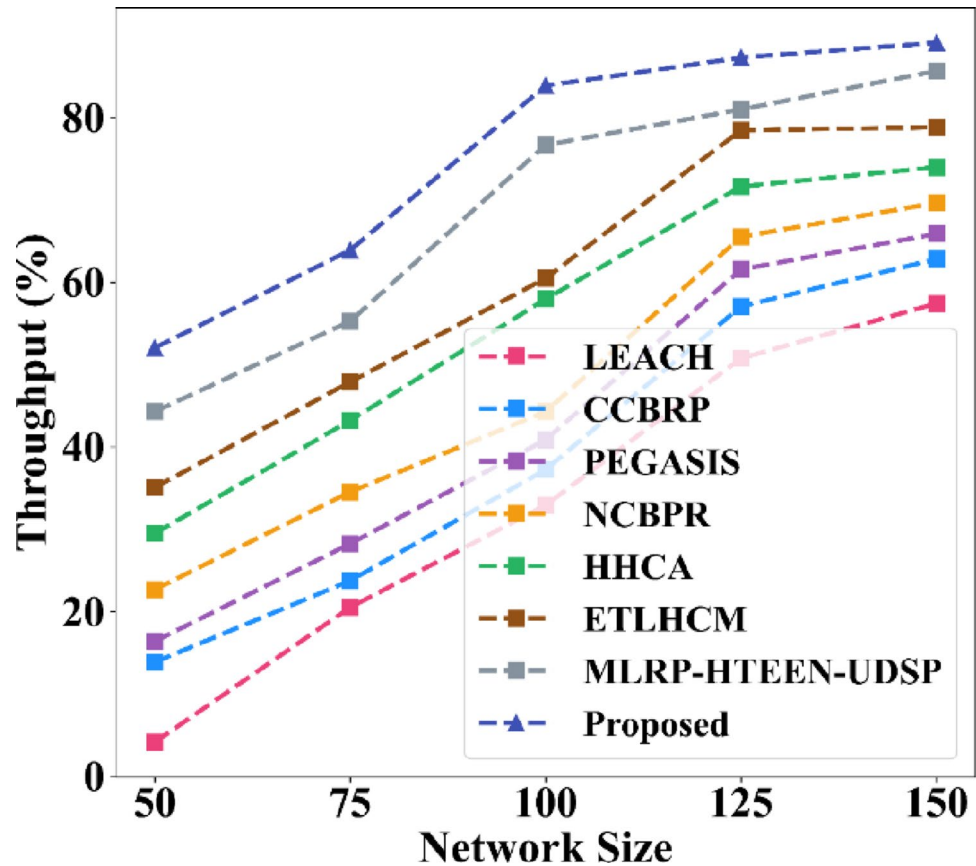


Table 14 Network lifetime of various models

Network Lifetime					
Techniques	50	75	100	125	150
LEACH	2.173	21.73	28.612	32.414	36.036
CCBRP	6.519	29.517	35.131	41.288	43.823
PEGASIS	10.865	33.863	42.012	46.901	49.618
NCBPR	17.565	41.288	53.239	55.775	64.467
HHCA	27.344	45.634	62.294	71.891	73.702
ETLHCM	34.044	50.161	70.805	76.962	80.765
MLRP-HTEEN-UDSP	40.563	54.869	75.875	79.859	85.473
Proposed	49.799	60.664	82.213	87.465	88.913

data delivery effectiveness. With a throughput of 52.02 at a modest network size of 50 nodes, the suggested model outperforms MLRP-HTEEN-UDSP with 44.28, ETLHCM with 35.1 and HHCA with 29.52. The suggested model continues to outperform the next best model, MLRP-HTEEN-UDSP, with a maximum throughput of 89.1 as the network grows to 150 nodes, while MLRP-HTEEN-UDSP records 85.68. The suggested model's optimal routing method, which reduces packet loss and increases dependable transmissions, is responsible for this constant advancement. The EE-BSAPO model guarantees reliable and high-throughput communication by properly employing energy-efficient pathways and taking trust and delay factors into account when choosing a route. This makes it ideal for real-time and data-intensive WSN applications. Table 15 represents throughput of the proposed and existing models.

The proposed EE-BSAPO protocol's Bit Error Rate (BER) performance was assessed and contrasted with that of other routing protocols, including LEACH, HEED, MBC, FRLDG, F-GWO and metaheuristics cluster related routing technique for energy efficient WSN (MHCRT-EEWSN) [33], at different node densities of 100, 200, 300,

Fig. 10 Throughput of the proposed and existing models**Table 15** Throughput for proposed and existing models

Throughput Techniques	50	75	100	125	150
LEACH	4.14	20.52	32.94	50.76	57.42
CCBRP	13.86	23.76	37.26	57.06	62.82
PEGASIS	16.38	28.26	40.86	61.56	65.88
NCBPR	22.68	34.56	44.28	65.52	69.66
HHCA	29.52	43.2	57.96	71.64	73.98
ETLHCM	35.1	47.88	60.48	78.48	78.84
MLRP-HTEEN-UDSP	44.28	55.26	76.68	81	85.68
Proposed	52.02	63.9	83.88	87.3	89.1

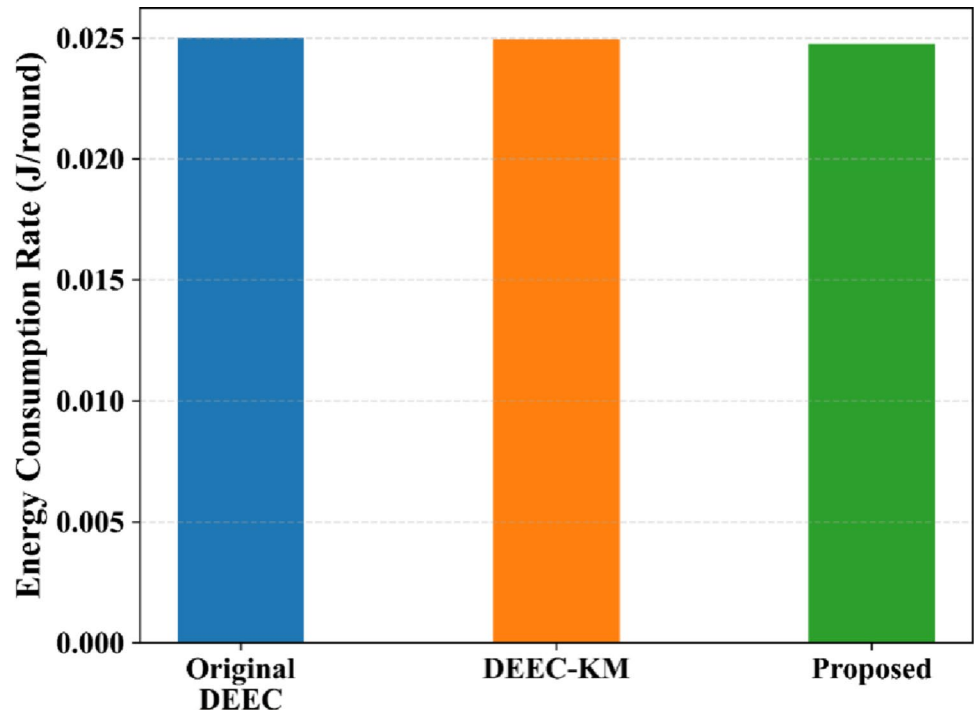
400 and 500. The suggested method perform improved performance with lowered BER values of 1.12 at 100 node, 1.75 at 200 nodes, 1.89 in 300 nodes, 3.16 at 400 nodes and 5.88 at 500 nodes respectively. The integration of the BSAPO method and enhanced fuzzy-based clustering system, which allow accurate route selection and handle reliable connection also enhance performance. Factors like distance, latency, residual energy, and trust are taken into account by EE-BSAPO to provide more dependable data transmission with low error propagation. This advantages increase communication quality and network longevity for energy-constrained and error-sensitive WSN applications. Table 16 represents the BER comparison of various models.

4.3 Performance Comparison with Various Existing Models

Figure 11 represents the energy consumption rate of the proposed and existing models. The existing models such as Original Distributed Energy-Efficient Clustering and DEEC- K-Means (DEEC-KM) [34] is used to compare

Table 16 BER comparison of various models

Number of nodes	LEACH	HEED	MBC	FRLDG	F-GWO	MHCRT-EEWSN	Proposed
100	12.75	9.63	5.82	4.46	2.56	1.20	1.12
200	15.06	10.71	7.05	5.96	4.33	2.02	1.75
300	18.05	14.92	11.66	8.00	6.37	2.02	1.89
400	24.16	18.46	14.38	11.36	9.22	4.87	3.16
500	26.06	24.84	19.41	15.06	11.12	7.59	5.88

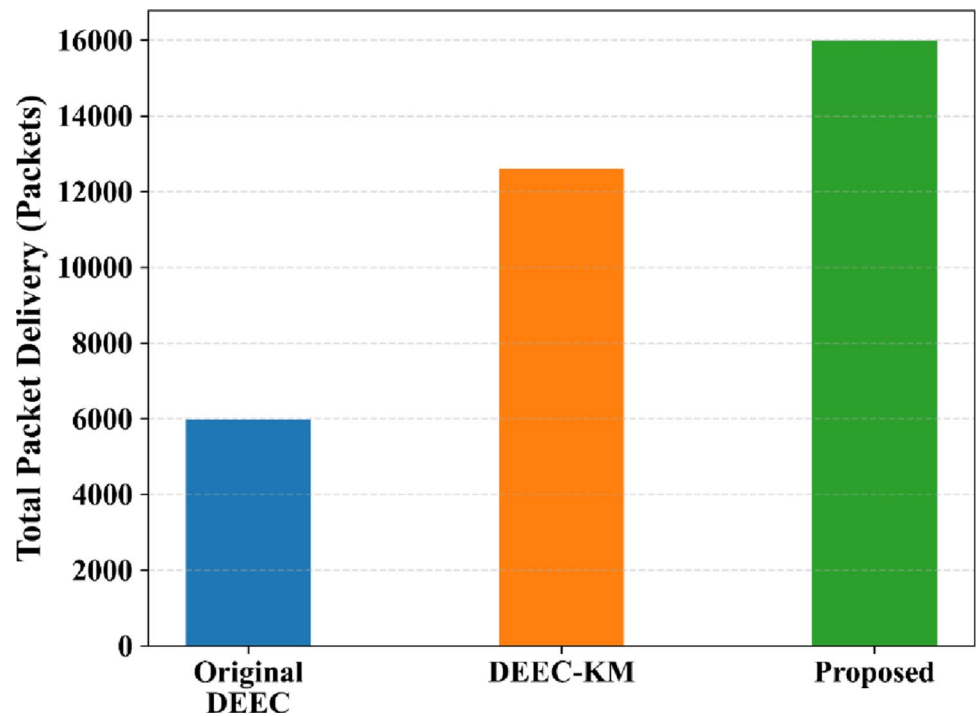
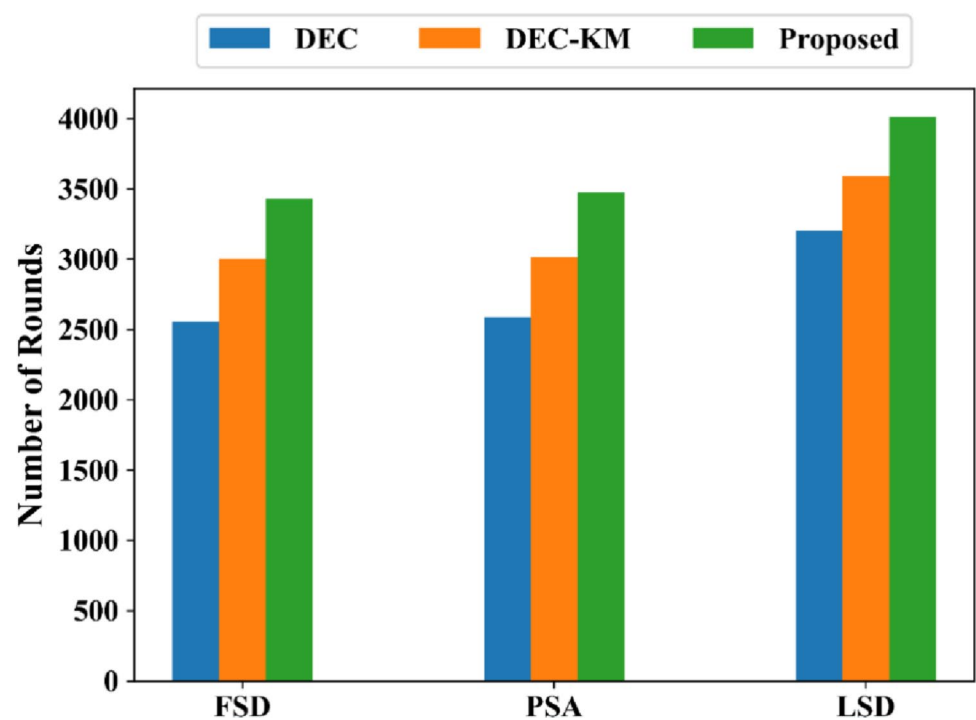
Fig. 11 Energy consumption rate of the proposed and existing models


with proposed model. The proposed model attained the energy consumption rate of 0.024773j/rounds which is better than the value of 0.024952j/round of DEEC-KM and 0.024997j/round of original DEEC, respectively. Figure 12 represents the total packet delivery of the proposed and existing models. The proposed model attained the total packet delivery of 15,984 packets which is way higher than the current models which obtained the value of 12,607 packets for DEEC-KM and 5982 packets for original DEEC, respectively.

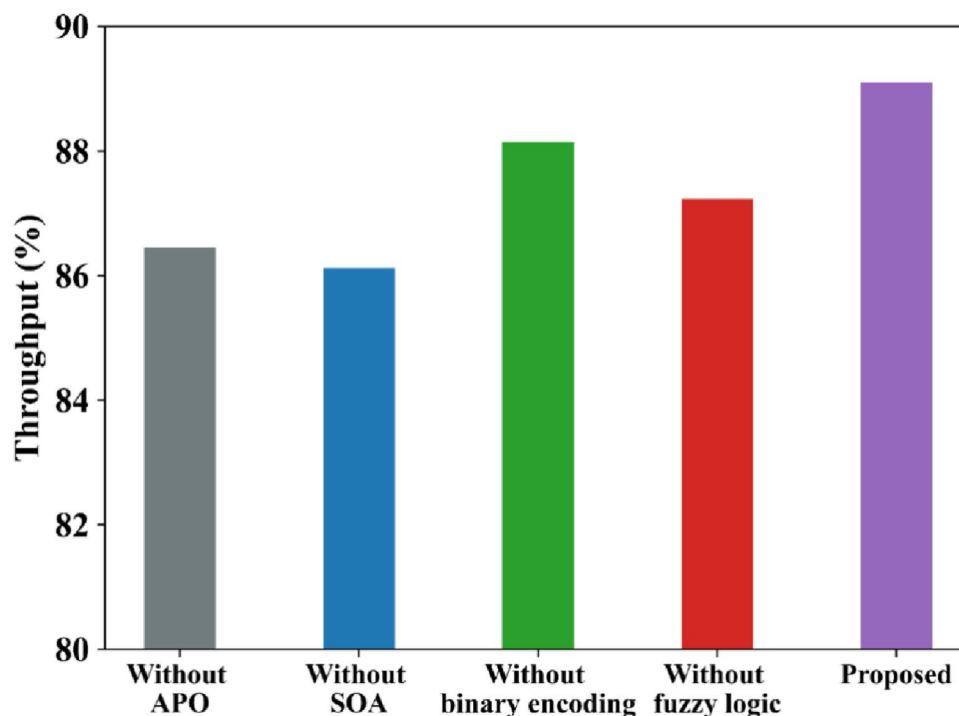
Figure 13 represents the first sensor node dies (FSD), last sensor node dies (LSD) and 90% of sensor nodes are still alive (PSA) of the proposed and existing models. The proposed model attained the FSD of 3428, PSA of 3475 and LSD of 4012, respectively. The proposed model attained the better results throughout the analysis by comparing with various existing models. Table 17 represents the FSD, PSA and LSD of the various models.

4.4 Ablation Study

The ablation study of the proposed model is given in the Fig. 14. Throughput is taken as the performance model in this analyse. The proposed model is analysed through five modules which the module 1 represents proposed model without APO, module 2 represents the proposed model without SOA and module 3 provides the proposed model without binary encoding. For the last two modules, the module 4 provides proposed model without fuzzy logic and module 5 represents the proposed model. The suggested model's ablation tests unequivocally demonstrate that every component makes a distinct contribution to overall performance and that eliminating any one of them results in a discernible reduction in routing efficiency. The optimizer's global search capability is lost in the

Fig. 12 Total packet delivery of various models**Fig. 13** FSD, PSA and LSD of the proposed and existing models**Table 17** FSD, PSA and LSD of the various models

Protocol	FSD	PSA	LSD
DEC	2554	2587	3200
DEC-KM [35]	3002	3011	3590
Proposed	3428	3475	4012

Fig. 14 Ablation Study

Table 18 Ablation study of the proposed model

Model Variant	Throughput (%)
Without APO	86.45
Without SOA	86.12
Without binary encoding	88.15
Without fuzzy logic	87.23
Proposed	89.1

absence of APO or SOA components, which causes early convergence and erratic routing pathways. Decisions about CH selection and link activation become more inaccurate and computationally demanding when binary encoding is excluded, forcing the system back into a continuous search space. Poor cluster formation results from the model's inability to adjust to real-time changes in energy, distance, latency and trust when fuzzy logic is eliminated. But when everything works together, binary encoding guarantees clear routing decisions, fuzzy logic selects high-potential nodes, and BSAPO (APO+Sculptor behaviour) effectively investigates and improves multi-hop pathways. The higher performance of the entire EE-BSAPO system may be explained by this synergy, which produces a routing mechanism that is simultaneously adaptive, fast-converging, and energy-balanced. Table 18 represents the values of the proposed model in each modules.

4.5 Computational Complexity Analysis

The proposed EE-BSAPO model provides considerable benefits by merging an enhanced fuzzy-based cluster-head selection mechanism with a binary sculptor-driven optimization approach that only works on the most relevant subset of nodes. EE-BSAPO cleverly narrows the search space with early fuzzy filtering, enabling only high-quality candidates to move on to the optimization step, in contrast to conventional APO, Binary-APO or SOA techniques that assess the whole network at each iteration. The dimensionality of route finding is reduced and superfluous calculations are reduced due to this selective processing. Furthermore, by directing protozoa agents toward discrete, energy-efficient routing choices rather than investigating unnecessary continuous states,

the binary sculptor transition mechanism speeds up convergence. Thus, the proposed EE-BSAPO model has attained the reduced complexity of $O(1)$. Because of this, the suggested model not only produces better energy balancing, trust-aware routing, and longer network lifetime, but it also achieves clearly reduced computing complexity, reaching near-linear behaviour with respect to network size. Compared to current fuzzy-metaheuristic clustering and routing approaches, EE-BSAPO is more scalable, quicker, and more appropriate for large-scale WSN installations because of its efficiency.

5 Conclusion

This study proposed a new novelty EE_BSAPO routing protocol for energy-conscious routing strategy. To effectively manage both optimum route finding and cluster formation, the model combines the Binary Sculptor Artificial Protozoa Optimization method with an enhanced fuzzy-based clustering mechanism. The protocol make intelligent decision-making in both the clustering and routing stages by grouping many factors, including residual energy, trust value, latency and distance. It improves the network's overall stability and efficiency. In-depth simulation-based analyses are used to comparing the proposed EE-BSAPO protocol against a number of advanced routing models. The results showed that EE-BSAPO in an essential performance metrics for current techniques. In specific, it operated with a remarkably low average end-to-end latency of 9.36ms, an average throughput of 75.24% and an average network lifetime of 73.81%. The model achieved an average energy efficiency of 72.32 J, for WSN with limited energy resources. In the future, more advanced CH selection strategies and adaptive routing methods are included to improve network's scalability and resilience. In order to detect and stop hostile node activity and guarantee safe and dependable data transfer across the network, a specialized trust evaluation and anomaly detection architecture will also be incorporated.

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Declarations

Competing interests The authors declare no competing interests.

Ethics, Consent to Participate, and Consent to Publish Not applicable.

Clinical Trial Not Applicable

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